Under What Conditions Does a Cross-Validated Peer Assessment Mechanism Induce Truthful Reporting and Eliminate Free-Riding in Student Group Work?

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Abstract

In a shared-reward production configuration that includes groups producing in higher education, the free-riding incentive is a major challenge. Institutional design, such as peer assessment review, seeks to mitigate this phenomenon. The main limitation of mechanisms such as the peer assessment factor is its sensitivity to collusion, especially reciprocal inflation, which effectively reduces the accuracy of such mechanisms. This project addresses this challenge by extending the peer assessment factor using a cross-validation methodology. The peer assessment factor uses consensus on a student's contribution as a reference point to evaluate the honesty of each rating and, from this, weights the credibility of each rating. The goal is to detect reciprocal inflation and adjust the credibility weight of each rating to reduce the effectiveness of collusion. This project uses mainstream peer assessment to make implementing such a solution feasible. Using numerical and Python simulations, this project proposes a cross-validated version of the peer assessment that makes reciprocal inflation less profitable while preserving a balanced budget.

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1 Introduction

The core configuration of a group project limits potential success to a Pareto-inferior outcome relative to the social optimum. In group work, the reward is shared, but the effort and cost of that effort are individual, which reduces the incentive to work as hard in group coursework as in individual coursework. The consequence of this configuration is an under-contribution by the team members, which constitutes the equilibrium. This phenomenon of under-contribution by most agents in a group is called free-riding: the best individual strategy is to under-contribute in a shared production. Holmström (1982) proved that no budget-balanced sharing rule among team members can implement first-best effort under moral hazard.

Group coursework is only an example of team production, so the question of how to maximise productivity in a team production where the reward is shared is an economic question applicable in many contexts.

Some mechanisms that mitigate the free-riding problem in teamwork include peer pressure and reputation concerns in repeated interactions (Kandel and Lazear, 1992). Repeated collaboration and mutual observability are required because they create the threat of future consequences that depend on knowing who contributed to what, and that is what makes peer pressure effective. One solution that has been tried in universities is peer assessment, which has been shown to be effective with the appropriate design (Topping, 1998; Brooks and Ammons, 2003). The main limitation of the peer assessment mechanism is that it is sensitive to collusion; even with the right design, peer assessment is biased by reciprocal inflation between collaborators (Brooks and Ammons, 2003; Magin, 2001).

Multiple designs solve the problems of free-riding and honest reporting. Li and Du (2025) use a mean-externality mechanism that is strategy-proof and collusion-proof for groups with at least three participants. This project uses a different approach: A cross-validated peer assessment factor (PAF) mechanism in which each rating's credibility weight is evaluated based on its deviation from the consensus of other peers. Unlike the mechanism of Li and Du (2025), this construction preserves budget balance and does not require structural changes to existing peer-assessment mechanisms, making it more institutionally tractable. It achieves the handling of both free-riding (via PAF) and collusion (via cross-validation).

Tho objective of this project is to study under what conditions does a cross-validated peer assessment mechanism induce truthful reporting and eliminate free-riding in student group work?

To answer this question, this project develops an analytical model complement to a python simulation. The analytical model develops three propositions, derives the equilibrium threshold $\alpha^* = 1/|\delta|$, and illustrates the result with a numerical example. The Python simulation validates the numerical example's results and serves as an interactive tool that allows users to vary the common grade and the penalty parameter.

The main contribution of this project is to address the potential collusion in the standard peer assessment model to address free-riding provides only a partial solution because peer assessment is sensitive to collusion. If this peer-assessment review is combined with a cross-validation

system, truthful reporting becomes a weak Nash equilibrium under unilateral deviation when $\alpha \geq 1/|\delta|$. The cross-validated peer assessment mechanism preserves budget balance: the original number of points attributed to the group does not change; the mechanism only affects the distribution.

Section 2 reviews the literature; Section 3 develops the model and the propositions; Section 4 reports the numerical results; and Section 5 concludes.

2 Literature Review

The free-rider problem in group projects is more than an anecdotal social issue; it is a challenge that undermines the maximisation of individual output in situations where collaborative results are shared and rarely attributable to specific members. The free-riding problem is a systematic under-contribution by group members due to the core structure of group work, which assigns a single grade to the entire team. Public goods are also vulnerable to free-riding, and solutions designed to modify incentives are frequently developed for one specific purpose and subsequently adapted for another. The literature on collaboration is extensive, but there are many factors that make the solution variable; conditions like repetition or the size of the group changes the potential solution. This review will analyse various solutions from the literature, such as peer pressure, and identify the weaknesses in the mechanisms used in group projects within the education sector.

2.1 Team Production and the Free-Rider Problem

The problem of free-riding arises from a simple analysis of the cost-reward calculations of each participant involved in a group project: since contributions are not individually rewarded and rewards are shared, players have every incentive to free-ride. Olson (1965) provided the theoretical foundation for this problem, demonstrating that voluntary contribution to a public good breaks down under rational self-interest. Yet his analysis also carries an optimistic implication for the present context: free-riding is more tractable in smaller groups. University students teams represent a scale at which well-designed incentive mechanisms stand a realistic chance of succeeding.

Holmström (1982) formalises the team production problem under a budget-balance sharing rule, for which the best effort of all team members cannot be implemented. He also establishes that equal grade-sharing in a group cannot achieve efficient effort.

To include those selective incentives, the first problem is the observability of effort, since individual effort cannot be observed through the production of joint output. Alchian and Demsetz (1972) examine the observability of effort in groups and conclude that observing each team member's effort would require an external monitor. One solution that requires lower cost would be to create a system in which team members both participate and monitor others.

Falkinger (1996) designs an incentive mechanism for public goods provision, which rewards or punishes team members as a function of their deviation from the group average. This mechanism

maintains budget balance and introduces efficient effort as the Nash equilibrium. This research is one of the main references on which the mechanism is based because of its use of the group average as a reference point.

The following section investigates whether informal social mechanisms within groups, specifically peer pressure, can serve as substitutes for formal contracts in enforcing effort.

2.2 Peer Pressure and Social Enforcement

The two keys to addressing free riding are mutual monitoring and a system of rewards; in the case of a private company, both are present through social mechanisms. As Kandel and Lazear (1992) demonstrated, peer pressure serves two functions in a firm. It acts as a monitoring device by imposing social sanctions on free riders and reinforces partnership by rewarding cooperative behaviour. However, Kandel and Lazear (1992) also specify that the conditions work only with repeated partnerships. Carpenter (2007) uses experimental evidence to show that peer punishment sustains cooperation in small groups, but it weakens in large groups. This claim is aligned with the proposition of Olson (1965). Bandiera et al. (2010) analyse workers' productivity and its dependence on social ties under different incentives, and according to this paper, under individual incentives, social ties reduce productivity, and under team incentives, social ties raise productivity. Peer pressure is a strong mechanism that effectively diminishes free riding. However, for this mechanism to be applicable, two conditions must be satisfied: perfect observability and repeated collaboration. Nonetheless, group projects in the educational sector rarely satisfy these conditions.

The subsequent section will explore which mechanisms are suitable for the educational format and examine the solutions attempted, assessing their success or failure.

2.3 Peer Assessment in Higher Education

The most common mechanism to reduce free riding in team work among students is peer assessment. At the end of a group project, each student evaluates the contributions and performance of their peers. One of the arguments in favour of the reliability of peer assessment is the high correlation between the teachers' grades and peer assessment in Topping's (1998) research.

The main design problems, according to the available literature, are the timing and the anonymity. Brooks and Ammons (2003) demonstrate that asking to grade the group's work at the end creates a bias of selective memory that tends to inflate grades and leads people to grade the entire group in the same way. A two-stage grading process would be more desirable and accurate. Anonymity is also very important because if peer assessment is public, others' judgments become a factor in the rating.

2.4 Collusion Failure

The discussion addresses collusion in peer assessment as stemming from the design and timing of the peer assessment. The potential failure of the mechanism due to strategic collusion has been addressed in the current literature, but there is no widely deployed peer assessment designed to be robust to it. Magin (2001) demonstrates, using an empirical study, that students who rate peers highly receive high ratings in return. The mechanism this project intends to build addresses this specific problem of collusion through reciprocity. Brooks and Ammons (2003) argue that collusion between two friends in the same group biases the mechanism, no matter how good the design features are. The project focuses on collusion rather than on developing the best design, since research on design features is extensive. Peer assessment that originally faced the free-riding problem now faces a problem of reporting honesty; the core reliability of the mechanism depends on how truthful the reporting is, and this is not ensured by the current mechanism that was reviewed.

2.5 Related Mechanism Design

This study revisits the main assessment method identified called PAF, which is adaptable to the cross-validation system and easily adjustable to address potential calibration issues. The main strength of this model is its self-correcting budget balance, which, rather than changing the total points assigned to the group, redistributes them more effectively. The PAF shifts the problem of free-riding to a matter of honest reporting. This is the contribution that this project aims to make to this gap in the literature. Li and Du (2025) agree with earlier work (Alon et al., 2011) on the claim that "preventing all forms of collusion is sometimes impossible" and propose the mean-externality mechanism as a solution for peer grading involving at least three candidates.

Overall, the PAF mechanism is vulnerable due to its heavy reliance on the honesty of group project members in assessing one another. To address the problem of collusion, this project implements a cost of dishonesty. The PAF mechanism is modified by including a cost for dishonest peer evaluation.

The mechanisms for truthful reporting in peer assessment systems include several key approaches. de Clippel et al. (2008) introduce impartial division mechanisms, where a player's outcome depends solely on reports from other participants. This mechanism inspired reliance on other students' grades to determine individual outcomes in group projects. Witkowski and Parkes (2012) present a peer prediction mechanism that encourages truthful reporting without requiring prior knowledge. Finally, Faltings and Radanovic (2017) offer a theoretical framework for credibility-weighted aggregation. These studies collectively demonstrate a research area focused on combining peer structure and cross-validation approaches that incorporate credibility weights.

Another possible mechanism that could have been used for this project is the Mean Externality Mechanism of Li and Du (2025). This alternative presents major differences that make it unsuitable for the context of this paper. The Mean Externality Mechanism requires individual expert assessments for each candidate, which limits its use compared with the PAF, which

only requires a common grade. The second characteristic that makes it less suitable for the education context is that it is not budget-balanced and requires additional input from teachers. However, the MEM is stronger in that it achieves full coalition-proofness, which is not the case for the PAF. Those two mechanisms have different strengths and limitations; choosing between them is a trade-off between theoretical reliability for the MEM and institutional feasibility and tractability for the PAF.

In conclusion, the existing PAF mechanism has been shown to mitigate free-riding but is highly sensitive to collusion and relies on the honesty of the report. Some mechanisms, such as the one proposed by Li and Du (2025), do not have those limitations, but they require structural modifications to the way group projects are conducted and assessed, making their application unlikely. This project extends the potential application of the PAF by adding some protection against reciprocal collusion and classic collusion, while preserving important characteristics that make it more likely to be used, such as budget balance.

The following methodology section describes the configuration and characteristics of the model to show formally its sensitivity to collusion and the effect of the cross-validation.

3 The Model

3.1 Setting up the Notation

Let us assume that n students must work on a group project. The teacher awards a grade G based on the group's joint output. Each student i chooses an effort level e_i, which incurs a cost $c(e_i)$, where $c(\cdot)$ is increasing and convex. The effort is bounded between:

$$e_i \in [0, e_{\max}] \tag{1}$$

where $e_{\max}$ is the maximum feasible effort level.

The joint output depends on each individual's effort, which results in the baseline grade:

$$G = f(e_1, \ldots, e_n) \tag{2}$$

where f is increasing in each e_i.

After the project, each student submits a rating from 1 to 5 on the contribution and the quality of their peers' work. Each student i submits:

$$r_{ij} \in \{1, \ldots, 5\} \tag{3}$$

Students rate only their peers; they cannot rate themselves, and ratings are anonymous to avoid retaliation.

Let θ_j denote the true contribution of student j, which is unobservable to the teacher but privately observable by peers who work alongside j. Because individual effort is unobservable to the professor, a free-riding problem arises.

Assumption 1 (Common knowledge)**.** *The structure of the game, the rating scale, and the grade formula are common knowledge among all students before the effort decisions are made.*

3.2 Stage 1 Baseline Game: Why Free-Riding Occurs

This section derives the free-riding result from the standard microeconomic foundation, which holds that students are self-interested and minimise their effort. In the context of a group project, the group produces a common output that leads to a common reward G. But this common output comes from individual effort, and each student i pays a personal cost $c(e_i)$ to contribute. Their payoff is the reward minus the effort cost:

$$u_i = \frac{G}{n} - c(e_i) \tag{4}$$

When a student decides to work harder, their individual share goes up by $1/n$ of that improvement. A marginal increase in student i's effort raises the group output by $\partial f/\partial e_i$, but student i only receives $(1/n) \cdot \partial f/\partial e_i$ of this output increase.

Each student maximises their payoff by choosing their effort level. Setting the derivative of u_i with respect to e_i equal to zero yields the individual's first-order condition:

$$\frac{1}{n}\frac{\partial f}{\partial e_i} = c'(e_i) \tag{5}$$

Define social welfare W as the total group output minus the total effort cost across all students. The social planner maximises W with respect to each e_i:

$$W = f(e_1, \ldots, e_n) - \sum_{i=1}^{n} c(e_i) \tag{6}$$

$$\frac{\partial W}{\partial e_i} = \frac{\partial f}{\partial e_i} - c'(e_i) = 0 \quad \Longrightarrow \quad \frac{\partial f}{\partial e_i} = c'(e_i) \tag{7}$$

According to the individual FOC, students stop working as soon as their marginal cost equals just a fraction $1/n$ of the marginal social benefit. Because $c'' > 0$, c' is strictly increasing in e_i, so the lower marginal-cost requirement in the individual FOC corresponds to a lower equilibrium effort than the social optimum. Every student puts in less effort than is optimal for the group. Holmström (1982) argues that budget-balanced team contracts (where the sum of points equals the baseline grade) cannot simultaneously provide first-best effort incentives, establishing the theoretical basis for why equal grade-sharing in group projects leads to systematic under-contribution.

The Nash equilibrium of this baseline game is therefore inefficient. Alchian and Demsetz (1972) propose monitoring individual effort, but this solution would be too costly for the university, which is why this paper turns to mutual monitoring among students. The challenge with mutual peer assessment is to build it such that the best strategy is honesty.

3.3 Stage 2: The PAF Baseline Mechanism

The PAF transforms the group grade into an individual grade by multiplying it by a peer-derived factor (calculations in Appendix I). The key property is balanced budget: the total number of points attributed to the group does not change; the distribution is simply adjusted to reflect the peer assessment ratings.

The grade of student j is no longer G/n. Under the PAF, the grade becomes:

$$G_j = G \times \frac{R_j}{\bar{R}} \tag{8}$$

where $R_j = (1/(n-1)) \sum_{i \neq j} r_{ij}$ is the average rating received by student j, and $\bar{R} = (1/n) \sum_j R_j$ is the group-average peer rating.

This change makes the grade individual. Since R_j depends on the effort exerted by student j, increased effort results in higher peer assessment scores. The payoff becomes:

$$u_j = G(e) \times \text{PAF}_j(e_j) - c(e_j) \tag{9}$$

where $\text{PAF}_j = R_j/\bar{R}$. The first-order condition is:

$$\frac{\partial G}{\partial e_j} \times \text{PAF}_j + G \times \frac{\partial \text{PAF}_j}{\partial e_j} = c'(e_j) \tag{10}$$

The marginal return on effort now exceeds $1/n$, so the student's optimal effort is higher than under equal splitting. The term $(\partial G/\partial e_j) \times \text{PAF}_j$ shows that individual contribution is now on a per-PAF scale rather than capped at $1/n$. The term $G \times (\partial \text{PAF}_j/\partial e_j)$ is the new incentive: extra effort increases the personal multiplier on the entire group grade.

Proposition 1. *The standard PAF mechanism is not collusion-proof. A coalition of two or more students can inflate each other's grades at the expense of honest students.*

Appendix I shows algebraically that collusion is profitable under the standard PAF in a worked example. The PAF shifts the problem of free-riding into a problem of collusion: because students can individually influence the grade, collusion among multiple agents becomes the new best strategy. The introduction of the cross-validation system aims to address the incentive to misreport.

3.4 The Cross-Validated PAF Mechanism

The concept is that if an individual's rating of a particular person significantly deviates from the consensus of others, that evaluation is discounted. When a student inflates a peer rating beyond the consensus defined by other students, the deviation creates an outlier that the cross-validated PAF mechanism penalises with respect to the calibration of α, where $\alpha \geq 0$ is the penalty parameter that scales how aggressively a deviation between a rater's report and the consensus is punished.

Li and Du (2025) addressed the possibility of combining peer assessment resistant to free riders with cross-validation resistant to collusion in groups of more than three agents. Their research does not use the PAF mechanism, however. The PAF mechanism is already used at certain universities (e.g., the University of Queensland), providing real-world institutional data. The goal of these systems is to achieve a better distribution of the total points already allocated to the group, rather than changing the overall sum set by the professor.

The intuition behind the credibility weight relies on the assumption that peers privately observe the effort others contribute to the group work. Under this assumption, if a rating deviates significantly from the consensus, less weight is assigned to it in the final outcome, treating it as an outlier.

To calibrate the model properly, the variable α is introduced as the penalty parameter, defining how harsh the penalty is. In this model, dishonesty is not costly for the rater of the dishonest peer review, but it is costly for the receiver.

Consensus estimate is defined as:

$$C_{ij} = \frac{1}{n-2} \sum_{\substack{k=1 \\ k \neq i, k \neq j}}^{n} r_{kj} \tag{11}$$

For student i's rating of student j, the consensus estimate is the average rating of j by all students except i and j.

Deviation:

$$d_{ij} = r_{ij} - C_{ij} \tag{12}$$

Students' ratings of another student might deviate from the consensus rating. The deviation from the consensus is just the distance between the individual rating r_{ij} and the consensus estimate C_{ij} of student j.

Credibility weight: Each individual rating receives a credibility weight which is defined as follows:

$$w_{ij} = \max(0,\, 1 - \alpha \cdot |r_{ij} - C_{ij}|) \tag{13}$$

where $\alpha \geq 0$ is the penalty parameter.

From (13) The Cross-validated weighted rating is defined as:

$$R_j^w = \frac{\sum_{i \neq j} w_{ij} \cdot r_{ij}}{\sum_{i \neq j} w_{ij}} \tag{14}$$

And the Cross validated PAF and individual grade are:

$$\mathrm{PAF}_j^w = \frac{R_j^w}{\bar{R}^w}, \qquad G_j = G \times \mathrm{PAF}_j^w, \qquad \bar{R}^w = \frac{1}{n}\sum_{j=1}^{n} R_j^w \tag{15}$$

Proposition 2 (Truthful Reporting as Weak Nash Equilibrium)**.** *For $\alpha \geq \alpha^*$, truthful reporting ($r_{ij} = \theta_j$ for all i, j) is a weak Nash equilibrium under unilateral deviation of the cross-validated PAF mechanism. The threshold α^* depends on the group size n and the rating scale.*

The derivation is provided in Section 4. By increasing α to an appropriate level, non-truthful reporting receives weight zero, which negatively impacts the grades of students who were over-rated. Setting α appropriately shifts the Nash equilibrium so that truthful reporting becomes the optimal solution.

Proposition 3 (Budget balance)**.** *The cross-validated PAF preserves budget balance:*

$$\sum_{j=1}^{n} G_j = n \times G \tag{16}$$

This follows directly from the definition: since $\mathrm{PAF}_j^w = R_j^w/\bar{R}^w$, the sum $\sum_{j=1}^{n} \mathrm{PAF}_j^w = n$, and therefore $\sum_{j=1}^{n} G_j = n \times G$.

4 Results

4.1 General Threshold

This section derives the general condition for truthful reporting before examining numerical examples. Consider that one player k is inflating his rating of student j by δ, while all other players are honest and do not deviate from the consensus. The consensus is equal to θ_j and the report is $\theta_j + \delta$. Since the consensus C_{kj} is built from all raters except k and j, and all other raters are honest:

$$C_{kj} = \theta_j \tag{17}$$

The deviation between rater k's report and the consensus is:

$$r_{kj} - C_{kj} = \theta_j + \delta - \theta_j = \delta \tag{18}$$

Applying the cross-validation weight formula:

$$w_{kj} = \max(0,\, 1 - \alpha|\delta|) \tag{19}$$

The dishonest rater's weight is zero when $1 - \alpha|\delta| \leq 0$, which gives:

$$\alpha^* = \frac{1}{|\delta|} \tag{20}$$

This provides a clear intuition: When a rater lies with a large deviation, the mechanism neutralises the report at relatively low values of α, since the weight $1-\alpha|\delta|$ falls quickly as $|\delta|$ grows. Conversely, small deviations require a larger α to be fully discounted. Overall, the higher the α, the less deviation the mechanism tolerates.

Table 1: Minimum α required to neutralise different lie sizes

Lie size $\lvert\delta\rvert$	Minimum α^*	Interpretation
1	1.00	Need maximum strictness
2	0.50	Moderate strictness sufficient
3	0.33	Low strictness sufficient
4	0.25	Minimal strictness sufficient

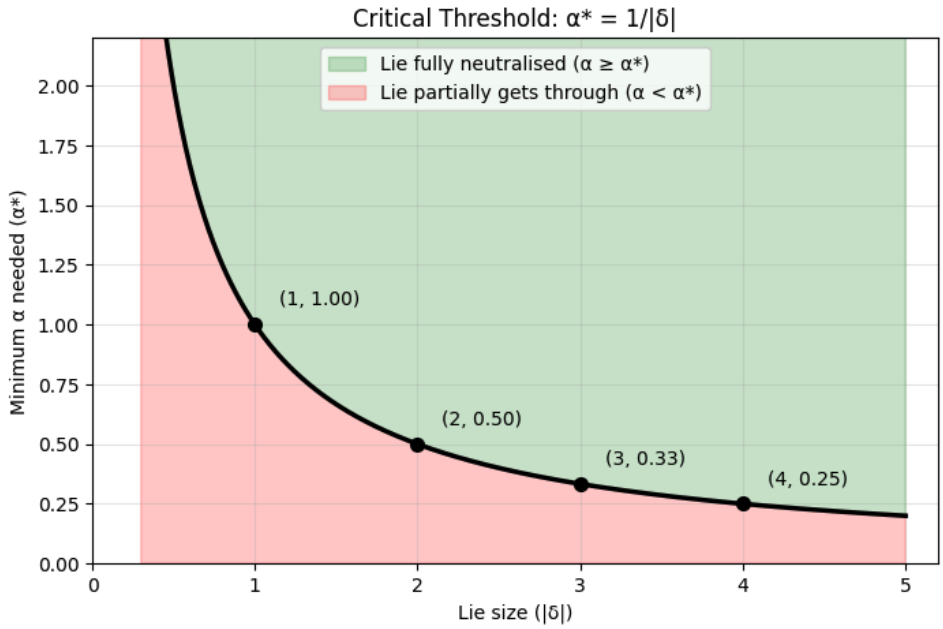


Figure 1: Critical threshold $\alpha^* = 1/|\delta|$. The green zone represents combinations where the lie is fully neutralised; the red zone where it partially gets through.

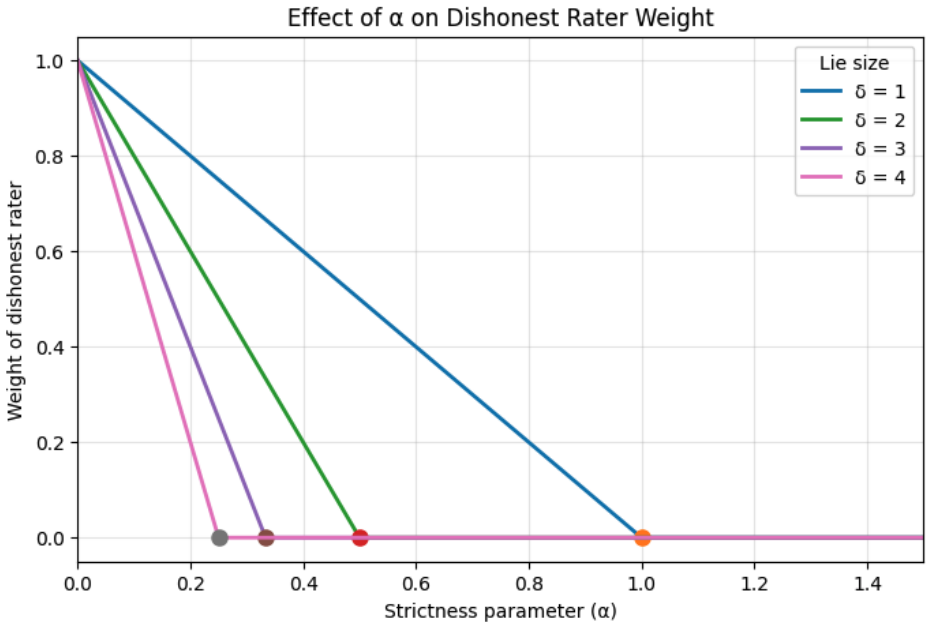


Figure 2: Effect of α on the dishonest rater's weight for different lie sizes. The dots mark the threshold α^* where the weight reaches zero.

This result indicates that α functions as a sensitivity control for the university: a higher α detects smaller lies but could also penalise individuals with different opinions. It serves as a tool that allows the university to set the acceptable level of deviation in peer assessments.

The threshold $\alpha^* = 1/|\delta|$ is derived under the assumption that exactly one student deviates and the rest report honestly. The mechanism neutralises the dishonest rating by reducing its weight to zero, but it does not penalise the dishonest rater's own grade, because the liar's PAF is defined by the ratings of the rest of the students who were honest. Truthful reporting is therefore a *weak* Nash equilibrium because a single dishonest person has no incentive to lie and will not be punished if they do. Furthermore, this result does not extend to simultaneous multi-rater collusion: if two or more students coordinate a deviation, the consensus estimate is affected because it is partly based on the dishonest report, so the threshold $\alpha^* = 1/|\delta|$ may no longer be sufficient. Analysing multi-rater collusion remains a direction for future research.

4.2 Numerical Example

Scenario. A group of 4 students: Alice (A), Bob (B), Chloé (C), and David (D). David is a free rider; Alice and Bob are friends who inflate each other's mutual rating. The collective group grade is $G = 72/100$.

Under perfect observability with honest reporting, $r_{ij} = \theta_j$ for honest raters. The true efforts are $\theta_A = 4$, $\theta_B = 3$, $\theta_C = 4$, $\theta_D = 2$. The collusion: Alice inflates her rating of Bob from 3 to 5 ($\delta_{AB} = 2$), and Bob inflates his rating of Alice from 4 to 5 ($\delta_{BA} = 1$). The inflations are asymmetric. All calculations are in Appendix I.

Table 2: Peer assessment with all players being honest

Rater \ Rated	Alice	Bob	Chloé	David
Alice	—	3	4	2
Bob	4	—	4	2
Chloé	4	3	—	2
David	4	3	4	—

Table 3: Peer assessment with Bob and Alice colluding

Rater \ Rated	Alice	Bob	Chloé	David
Alice	—	5 (inflated, +2)	4	2
Bob	5 (inflated, +1)	—	4	2
Chloé	4	3	—	2
David	4	3	4	—

Table 4: Comparison of grades across mechanisms

Student	PAF honest	Standard PAF (collusion)	Cross-Val $\alpha = 0.5$	Cross-Val $\alpha = 0.99$
Alice	88.6	89.1	92.4	88.8
Bob	66.5	77.5	65.2	66.4
Chloé	88.6	82.3	86.9	88.6
David	44.3	41.1	43.5	44.3

Standard PAF under collusion. Under standard PAF with collusion, Alice's grade rises from 88.6 to 89.1 (+0.5); Bob's rises from 66.5 to 77.5 (+11.0). The bigger lie pays the bigger return: Bob, who was inflated by $\delta = 2$, gains substantially more than Alice, who was inflated by $\delta = 1$. Chloé and David's grades fall by 6.3 and 3.2 points respectively. This confirms Proposition 1: standard PAF is not collusion-proof, and the cost of collusion is borne by the honest students.

Cross-validated PAF with $\alpha = 0.5$. At $\alpha = 0.5$, the larger inflation ($\delta = 2$) is fully discounted: $w_{AB} = \max(0, 1 - 0.5 \cdot 2) = 0$. The smaller inflation ($\delta = 1$) is only half-discounted: $w_{BA} = 0.5$. The result is an *asymmetric coalition outcome*: Alice's grade rises to 92.4 (+3.8 vs honest baseline), while Bob's grade falls to 65.2 (−1.3 vs honest baseline). The coalition is internally unstable. Bob would refuse to participate if he knew the result.

Cross-validated PAF with $\alpha = 0.99$. As α approaches $1 = 1/\delta_{\min}$, both inflations are essentially fully discounted. All four students return to within 0.2 of their honest baseline grades. The joint coalition gain collapses to approximately +0.1 points across the pair —

essentially zero. Note that $\alpha = 0.99$ is used in place of the limit $\alpha \to 1^-$ to keep all credibility weights strictly positive; at $\alpha = 1$ exactly, the corrupted consensus would also discount Chloé's and David's honest ratings to zero, leaving Bob's weighted rating undefined.

Chloé's grade trajectory is 88.6 (honest) $\to$ 82.3 (PAF + collusion) $\to$ 86.9 ($\alpha = 0.5$) $\to$ 88.6 ($\alpha = 0.99$). David's trajectory is 44.3 $\to$ 41.1 $\to$ 43.5 $\to$ 44.3. Honest students suffer under unmitigated collusion; cross-validation restores them toward their honest baselines as α increases. Budget balance holds throughout (every column sums to $4 \times 72 = 288$).

Three economic findings. The asymmetric numerical example produces three findings beyond the unilateral analysis of Section 4.1:

1. **Coalition internal instability.** At $\alpha = 0.5$, the larger lie is fully discounted but the smaller lie is not. Alice gains; Bob loses. Without side payments or social ties, the coalition is not stable.

2. **The bigger liar is punished first.** Larger deviations cross the zero-weight threshold at lower α. As α rises from 0, the mechanism neutralises the largest lie first.

3. **Mechanism stability ceiling.** At $\alpha = 1/|\delta_{\min}|$ in this scenario, honest raters' weights also collapse because the corrupted consensus places them one unit from the apparent consensus. Practical implementation requires α strictly below this ceiling.

4.3 Simulation Results

To analyse and test the model under different conditions, this paper includes a Python simulation in Appendix II. The simulation is run on Google Colab and reuses the same context as the numerical example. It has an interactive slider to adjust α and an interactive metric to adjust the professor's assigned base grade (Figure 3). It produces three visualisations: a bar chart comparing grades under the standard PAF and cross-validation configurations (Figure 4), a line chart showing how each student's grade evolves as α changes (Figure 5), and a heatmap of credibility-weighted values (Figure 6).



Figure 3: Interactive simulation controls: α slider and group-grade input.

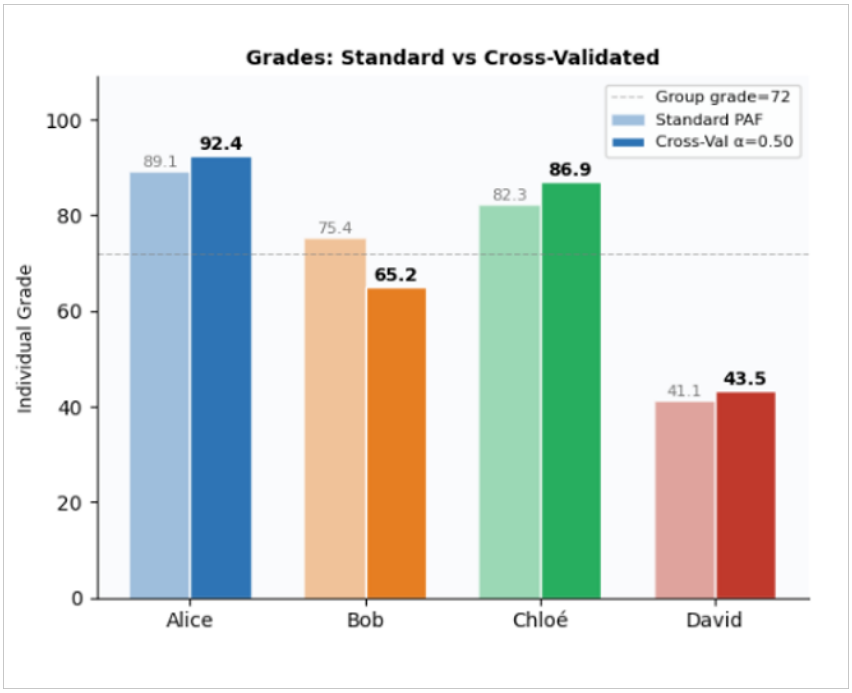


Figure 4: Individual grades under standard PAF vs. cross-validated PAF at $\alpha = 0.50$.

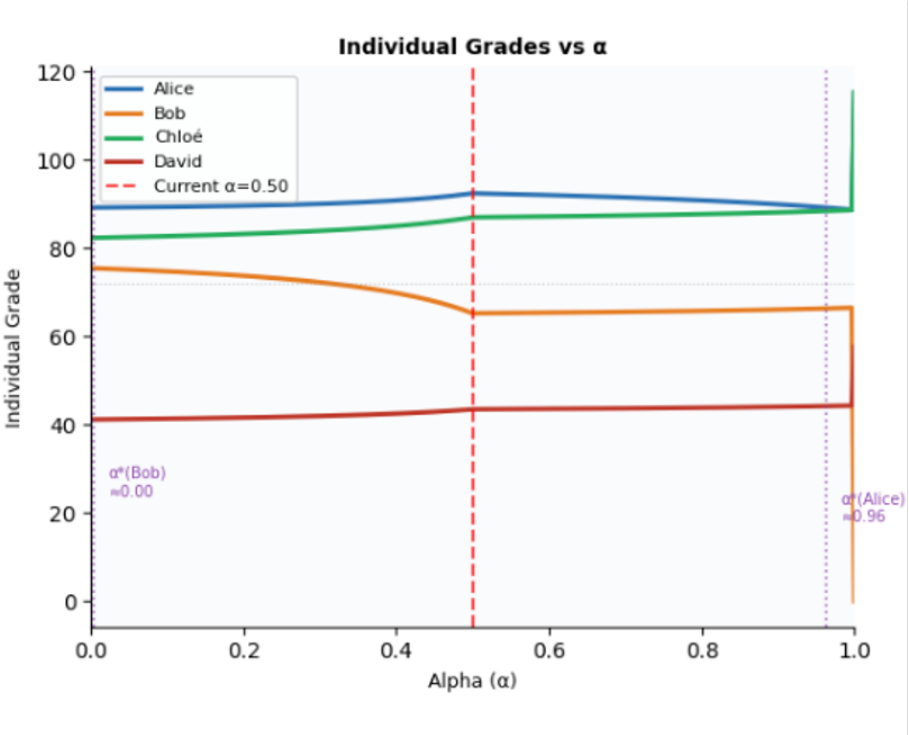


Figure 5: Individual grades as a function of α. Dashed line: current $\alpha = 0.50$. Dotted line: $\alpha^*(\text{Alice}) \approx 0.96$.

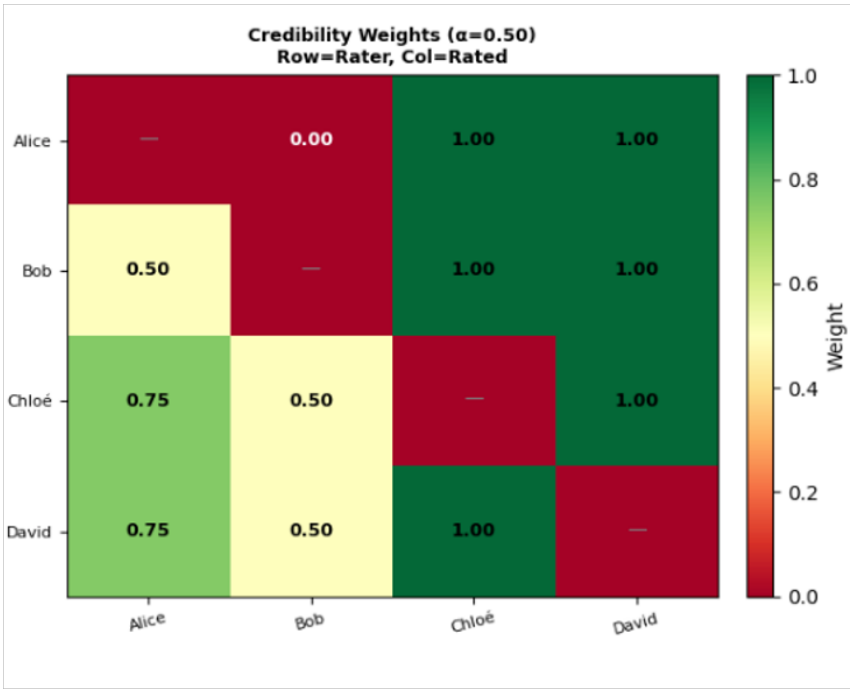


Figure 6: Credibility weights w_{ij} at $\alpha = 0.50$. Rows: raters. Columns: rated.

```
================================================================================
  Student       Std PAF    Cross-Val   Difference  Colluder?
================================================================================
  Alice            89.1         92.4         +3.2      ⚠ YES ←
  Bob              75.4         65.2        -10.2      ⚠ YES ←
  Chloé            82.3         86.9         +4.7
  David            41.1         43.5         +2.3
================================================================================
  Budget check: 288.0 = 4 × 72 = 288  ✓

  ⚡ KEY INSIGHT: At α=0.50, collusion BACKFIRES for Bob
     Bob loses 10.2 points compared to honest reporting
     → Rational students would NOT collude at this α level
```

Figure 7: Simulation output: numerical comparison of grades at $\alpha = 0.50$ with budget-balance and collusion-backfire summary.

The interactive slider that changes the value of α enables observation that, at sufficiently high α, collusion is no longer a profitable strategy. For a low α, there is very little difference from the standard PAF. As α increases, Bob's grade decreases and Chloé's increases proportionally. This shows the specific α from which collusion is a losing strategy. This α depends on the game's configuration; it is not constant. The credibility-weight heatmap shows that at high α, Alice's rating of Bob and Bob's rating of Alice are weighted near zero; the mechanism ignores their inflated ratings.

The simulation confirms the theoretical prediction of Proposition 2. The simulation is useful for further research to analyse rating data and determine how to calibrate α. The mechanism code is available in Appendix II.

5 Conclusion

In this project, we have shown that the standard PAF mechanism is not collusion-proof: a coalition of two or more students can inflate each other's grades at the expense of honest students. For $\alpha \geq \alpha^*$, truthful reporting ($r_{ij} = \theta_j$ for all i, j) is a weak Nash equilibrium under unilateral deviation of the cross-validated PAF mechanism.With alpha that is the penalty parameter that scales how aggressively a deviation between a rater's report and the consensus is punished. The threshold α^* depends on the group size n and the rating scale. Adding the cross-validation system maintains budget balance: the mechanism redistributes the same total no marks are created or destroyed.

The cross-validated peer assessment mechanism eliminates free-riding and induces truthful reporting when: (i) the penalty parameter exceeds the threshold, $\alpha \geq \alpha^*$; (ii) the group size satisfies $n \geq 4$; (iii) effort is mutually observable, which implies that the consensus C_{ij} is meaningful. The mechanism preserves budget balance by construction. The Python simulation confirmed the threshold behaviour in the worked example. The gains from the coalition of Bob and Alice are positive for $\alpha = 0.5$ (+2.5 points overall), but Alice gains +3.8 and Bob loses 1.3 the coalition becomes unstable. For $\alpha = 0.99$, the joint gain collapses to +0.1 points. This shows that there is no universal α that makes collusion backfire; rather, for each scenario, there is an α value that makes collusion inefficient.

In the simulation, Alice, who colludes with Bob, does not see her grade reduced by inflating Bob's grade. This is the main limitation of the model: the mechanism makes lying inefficient in proportion to the lie, so a player with low individual deviation is not directly punished. However, the fact that the coalition backfires on Bob makes a coalition between these two players in this specific context unlikely. If the model is known, Bob will not join a coalition that lowers his grade. This warrants further investigation: if one member of the coalition gains no benefit from it, the act of collaboration may be deterred. Assuming that (i) there are more truth-tellers than liars, (ii) at least one liar has their grade penalised by the system, and (iii) the teacher sets α such that the coalition is inefficient, then liars who realise they will be penalised will refrain from joining the coalition, triggering a backward induction that leaves the coalition empty.

Overall, the combination of the established PAF with a cross-validated credibility penalty reduces the collusion vulnerability identified by Brooks and Ammons (2003) and connects to the mean-externality framework of Li and Du (2025).

The mechanism has limitations that must be acknowledged. The first comes from the assumption that students can perfectly observe the amount of work other students put in; in reality, those observations can be noisy. The second is that this is a one-shot game; it does not account for the implications of repeated interactions. The final limitation is the lack of consideration for potential differences in risk behaviour and broader behavioural differences among students, which could be addressed by incorporating experimental observations through surveys. Past research by Brooks and Ammons (2003) demonstrates that peer assessment surveys including multiple rounds during the project are more accurate.

The main strength of the model is its potential applicability at the university. A pilot phase

could be conducted that would not impact student grades, with the cross-validation system run in parallel with the PAF. This would calibrate α empirically using observed rating distributions. The cohort grade average and the total number of points would not change, ensuring that the value attributed to the work is not altered.

Further research is needed to improve the model; a survey and some real-world experiments would help make it more accurate. An experiment could be conducted with three conditions: (1) a one-shot standard peer assessment baseline to benchmark unmitigated misreporting; (2) a repeated standard peer assessment across multiple rounds to observe if peer accountability reduces misreporting; (3) a one-shot cross-validated condition to evaluate whether its misreporting levels match the repeated condition. This would test whether institutional design can substitute for repeated interaction in generating accountability.

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Appendix I: Mathematical Proofs

Scenario. Group of 4 students: Alice (A), Bob (B), Chloé (C), David (D). David is a free rider; Alice and Bob are friends who inflate their mutual ratings. Group grade $G = 72/100$.

Scenario 1: Standard PAF (no cross-validation), honest ratings

Step 1 — Average rating:

$$R_A = \tfrac{4+4+4}{3} = 4.00, \qquad R_C = \tfrac{4+4+4}{3} = 4.00$$
$$R_B = \tfrac{3+3+3}{3} = 3.00, \qquad R_D = \tfrac{2+2+2}{3} = 2.00$$

Step 2 — Group average:

$$\bar{R} = \tfrac{4.00+3.00+4.00+2.00}{4} = 3.25$$

Step 3 — PAF and individual grades:

$$\text{PAF}_A = \tfrac{4.00}{3.25} = 1.23, \qquad G_A = 72 \times 1.23 = 88.6$$
$$\text{PAF}_B = \tfrac{3.00}{3.25} = 0.92, \qquad G_B = 72 \times 0.92 = 66.5$$
$$\text{PAF}_C = \tfrac{4.00}{3.25} = 1.23, \qquad G_C = 72 \times 1.23 = 88.6$$
$$\text{PAF}_D = \tfrac{2.00}{3.25} = 0.61, \qquad G_D = 72 \times 0.61 = 44.3$$

Budget check: $88.6 + 66.5 + 88.6 + 44.3 = 288 = 72 \times 4$ ✓

Scenario 2: Standard PAF with collusion

Step 1 — Average rating:

$$R_A = \tfrac{5+4+4}{3} = 4.33, \qquad R_C = \tfrac{4+4+4}{3} = 4.00$$
$$R_B = \tfrac{5+3+3}{3} = 3.67, \qquad R_D = \tfrac{2+2+2}{3} = 2.00$$

Step 2 — Group average:

$$\bar{R} = \tfrac{4.33+3.67+4.00+2.00}{4} = 3.50$$

Step 3 — PAF and individual grades:

$$G_A = 72 \times \tfrac{4.33}{3.50} = 89.1$$
$$G_B = 72 \times \tfrac{3.67}{3.50} = 77.5$$
$$G_C = 72 \times \tfrac{4.00}{3.50} = 82.3$$
$$G_D = 72 \times \tfrac{2.00}{3.50} = 41.1$$

Budget check: $89.1 + 77.5 + 82.3 + 41.1 = 288$ ✓

Scenario 3a: Cross-validation PAF, $\alpha = 0.5$ (with collusion)

Weights for ratings of Alice ($j = A$):

$r_{BA} = 5,$	$C_{BA} = \frac{4+4}{2} = 4,$	$d = 1,$	$w_{BA} = \max(0, 1 - 0.5 \cdot 1) = 0.5$
$r_{CA} = 4,$	$C_{CA} = \frac{5+4}{2} = 4.5,$	$d = -0.5,$	$w_{CA} = 0.75$
$r_{DA} = 4,$	$C_{DA} = 4.5,$	$d = -0.5,$	$w_{DA} = 0.75$

Weights for ratings of Bob ($j = B$):

$r_{AB} = 5,$	$C_{AB} = \frac{3+3}{2} = 3,$	$d = 2,$	$w_{AB} = \max(0, 1 - 0.5 \cdot 2) = 0$
$r_{CB} = 3,$	$C_{CB} = \frac{5+3}{2} = 4,$	$d = -1,$	$w_{CB} = 0.5$
$r_{DB} = 3,$	$C_{DB} = 4,$	$d = -1,$	$w_{DB} = 0.5$

For Chloé and David, all weights = 1 (honest raters agree on the rating).

Weighted average ratings:

$$R_A^w = \tfrac{0.5\times5+0.75\times4+0.75\times4}{0.5+0.75+0.75} = \tfrac{8.5}{2.0} = 4.25$$
$$R_B^w = \tfrac{0\times5+0.5\times3+0.5\times3}{0+0.5+0.5} = \tfrac{3}{1} = 3.00$$
$$R_C^w = 4.00, \quad R_D^w = 2.00$$
$$\bar{R}^w = \tfrac{4.25+3.00+4.00+2.00}{4} = 3.3125$$

Individual grades:

$$G_A = 72 \times \tfrac{4.25}{3.3125} = 92.4$$
$$G_B = 72 \times \tfrac{3.00}{3.3125} = 65.2$$
$$G_C = 72 \times \tfrac{4.00}{3.3125} = 86.9$$
$$G_D = 72 \times \tfrac{2.00}{3.3125} = 43.5$$

Scenario 3b: Cross-validation PAF, $\alpha = 0.99$ (with collusion)

Weights for ratings of Alice:

$$w_{BA} = \max(0, 1 - 0.99 \cdot 1) = 0.01$$
$$w_{CA} = \max(0, 1 - 0.99 \cdot 0.5) = 0.505$$
$$w_{DA} = 0.505$$

Weights for ratings of Bob:

$$w_{AB} = \max(0, 1 - 0.99 \cdot 2) = 0$$
$$w_{CB} = \max(0, 1 - 0.99 \cdot 1) = 0.01$$
$$w_{DB} = 0.01$$

Weighted average ratings:

$$R_A^w = \frac{0.01\times5+0.505\times4+0.505\times4}{1.02} = \frac{4.09}{1.02} = 4.01$$
$$R_B^w = \frac{0\times5+0.01\times3+0.01\times3}{0.02} = \frac{0.06}{0.02} = 3.00$$
$$R_C^w = 4.00, \quad R_D^w = 2.00$$
$$\bar{R}^w = \frac{4.01+3.00+4.00+2.00}{4} = 3.2525$$

Individual grades:

$$G_A = 72 \times \frac{4.01}{3.2525} = 88.8$$
$$G_B = 72 \times \frac{3.00}{3.2525} = 66.4$$
$$G_C = 72 \times \frac{4.00}{3.2525} = 88.6$$
$$G_D = 72 \times \frac{2.00}{3.2525} = 44.3$$

Full comparison table

Table 5: Comparison of grades across mechanisms (Appendix I summary)

Student	PAF honest	Standard PAF + collusion	Cross-Val $\alpha = 0.5$	Cross-Val $\alpha = 0.99$
Alice	88.6	89.1	92.4	88.8
Bob	66.5	77.5	65.2	66.4
Chloé	88.6	82.3	86.9	88.6
David	44.3	41.1	43.5	44.3

Appendix II: Python Simulation

The Python implementation of the cross-validated PAF mechanism is available in the project's Google Colab notebook. The notebook provides an interactive slider for α and a numeric input for the baseline group grade. It generates the bar chart, line chart, and credibility-weight heatmap shown in Section 4.3.

The core computation function is reproduced below for transparency.

```
import numpy as np

def cross_validation_grades(n, ratings, alpha, G=72.0):
    """
    Compute individual grades under cross-validated PAF.
    n        : group size
    ratings  : n x n matrix (np.nan on diagonal)
    alpha    : penalty parameter
    G        : per-student baseline grade
    """
    R_w = np.zeros(n)
    for j in range(n):
        weighted_sum = 0.0
        weight_sum = 0.0
        for i in range(n):
            if i == j:
                continue
            others = [k for k in range(n) if k != i and k != j]
            C_ij = np.mean([ratings[k, j] for k in others])
            d_ij = ratings[i, j] - C_ij
            w_ij = max(0.0, 1.0 - alpha * abs(d_ij))
            weighted_sum += w_ij * ratings[i, j]
            weight_sum += w_ij
        if weight_sum > 0:
            R_w[j] = weighted_sum / weight_sum
        else:
            R_w[j] = np.nanmean([ratings[i, j] for i in range(n) if i != j])
    R_avg = np.mean(R_w)
    PAF_w = R_w / R_avg
    return G * PAF_w
```

The visualisation code (matplotlib bar chart, line chart, heatmap) is documented in the project repository.

Appendix III: AI Usage Declaration

This appendix documents the use of AI tools during the preparation of this project, in accordance with the EC3009 module guidelines on the permitted use of artificial intelligence.

1. Tools Used

The following AI tools were used during the preparation of this project:

- Claude (Anthropic), accessed via Cowork mode during March to May 2026.

2. Permitted Uses of AI

AI tools were used for the following functions, all of which fall within the categories permitted by the EC3009 guidelines:

- **Feedback on writing structure.** Sentence-level critique, identification of weak arguments, suggested placements for citations, flagging of undefined notation, identification of redundant content, and structural feedback on paragraph organisation in the introduction, literature review, methodology, results, and conclusion sections.
- **LaTeX conversion and typesetting.** Conversion of the manuscript from a Microsoft Word draft to LaTeX, including math typesetting, table formatting using the `booktabs` package, bibliography formatting in the requested house style, and resolution of cross-references between body and appendices.
- **Interactive Python visualisation code.** See Section 4 of this appendix for the detailed declaration regarding the simulation.

3. Declaration Regarding the Interactive Simulation

The Python simulation provided in Appendix II involves two distinct components, declared separately below.

Component A Mechanism logic (student's own work). The student designed the simulation's logical structure, including:

- The implementation of the cross-validation formula
- The weight computation
- The cross-validated PAF and individual grade computation
- The comparison structure between standard PAF (no cross-validation) and the cross-validated mechanism at varying values of α

The student verified that these computations correctly reproduce the analytical results derived in Section 3 and the worked example in Appendix I.

Component B Interactive visualisation layer (AI-assisted). The interactive visualisation components were generated with Claude AI assistance. Specifically:

- The Jupyter `ipywidgets` slider for the penalty parameter α and the numeric input for the baseline group grade G.
- The `matplotlib` bar chart comparing grades under standard PAF and cross-validated PAF (Figure 4).
- The `matplotlib` line chart showing individual grade evolution as α varies (Figure 5).
- The `matplotlib` heatmap of credibility weights w_{ij} (Figure 6).
- The formatted simulation output table (Figure 7).

AI assistance was used for the visualisation code because the student's training in this project is in the economics of mechanism design rather than in Python visualisation libraries. The student specified the desired visualisations (what to display, what to vary, what colour scheme to use) and verified that each output correctly represented the mechanism described in Sections 3 and 4. The student has read the visualisation code and understands its function.

4. Sample Prompts and Outputs

The following examples illustrate representative uses of AI during this project. They are presented in the order in which they were used during the writing process.

Example 1 — Structural feedback on a paragraph. *Prompt:*

> This is my introduction paragraph for the literature review. What do you think? Can you tell me what to cut, what to keep, and what is missing?

AI response (excerpt):

> *The major structural elements are present: structural problem stated, public-goods generalisation, preview at the end, no rhetorical questions. Two issues to address. First, "I will analyze various solutions" is first-person voice — switch to "This review analyses..." to stay consistent with the rest of the paper. Second, the order of ideas is slightly off: the sharp definition of free-riding should appear before the public-goods generalisation, not after. Reorder so the trajectory reads: define the problem → generalise to other settings → preview the review's structure.*

Example 2 — Verification of arithmetic. *Prompt:*

> Can you check the arithmetic in my Appendix I and tell me if the numbers in the comparison table are consistent with the step-by-step calculations?

AI response (excerpt):

> *The body of Appendix I derives the honest baseline as Alice* = 88.6*, Bob* = 66.5*, Chloé* = 88.6*, David* = 44.3*. However, the summary comparison table reports Alice* = 83.4 *and Bob* = 75.7 *for the honest baseline. These numbers are inconsistent with the body calculations. Verify which set is canonical (the body calculations are correct under the perfect-observability assumption) and update the summary table accordingly.*

Example 3 — Python visualisation code generation. *Prompt:*

> Can you write Python code for a bar chart that compares four students' individual grades under standard PAF and cross-validated PAF, with the group grade shown as a horizontal reference line?

AI response (excerpt — code):

```
import matplotlib.pyplot as plt
import numpy as np

students = ['Alice', 'Bob', 'Chloé', 'David']
standard_paf = [89.1, 77.5, 82.3, 41.1]
cross_val   = [92.4, 65.2, 86.9, 43.5]

x = np.arange(len(students))
width = 0.35

fig, ax = plt.subplots(figsize=(8, 5))
ax.bar(x - width/2, standard_paf, width, label='Standard PAF')
ax.bar(x + width/2, cross_val,   width, label='Cross-Val =0.50')
ax.axhline(72, color='gray', linestyle='--', label='Group grade=72')
ax.set_xticks(x); ax.set_xticklabels(students)
ax.set_ylabel('Individual Grade')
ax.legend()
plt.show()
```

Example 4 — LaTeX formatting assistance. *Prompt:*

> How do I format the appendix headings so they read "Appendix I:" and "Appendix II:" instead of "Appendix A:" and "Appendix B:"?

AI response (excerpt):

> *Use starred sections to bypass automatic numbering, and hard-code the appendix labels in the section title. Replace* `\section{Mathematical Proofs}` *with* `\section*{Appendix I: Mathematical Proofs}` *and add* `\addcontentsline{toc}{section}{Appendix I: Mathematical Proofs}` *immediately below to keep it in the table of contents. Repeat for Appendix II. After this change, internal cross-references using* `\ref{}` *will no longer resolve; replace those with hard-coded text such as "Appendix I" or "Appendix II".*

All written content is the student's own except where declared.The student has reviewed all AI outputs and understood them.No AI text has been copied into the body without rewriting

Signed: John Headley *Date:* May 17, 2026